KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 9 · SCIENCE

The Solar System and Stars

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. Name as many planets as you can. Try to put them in order from the Sun.

2. What do you think keeps the planets moving around the Sun?

3. Is the Sun a star? Explain your guess.

4. Draw a pattern of stars you have seen, or invent one. Give it a name and tell what it shows.

5. What is one thing you wonder about space?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Explain how gravity and a planet's forward motion work together to make an orbit.

2. Use the planet data table: describe the pattern between distance from the Sun and the length of a year. Give two planets as examples.

3. Why does the Sun look so much bigger and brighter than the other stars?

4. Choose two cultures from today's lesson. How did each one read the sky, and what did they use that knowledge for?

★ **Challenge:** On a clear night, with a parent, use the Big Dipper's pointer stars to find Polaris. Draw what you saw and mark which way is north.

Looking Ahead: Next week is the Unit 2 Science Checkpoint, covering landforms, the water cycle and weather, Earth, Sun and Moon, and the solar system. Question to bring: How does gravity connect a falling raindrop, the Moon's orbit and the planets?